# EdGame Analytics Platform - Progress Report

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## Executive Summary

The EdGame Analytics Platform has reached a major milestone: **all five educational games are now fully implemented** as playable KAPLAY.js browser games. Each game embeds stealth assessment through Evidence-Centered Design (ECD), measuring student competencies across six analytics dimensions without interrupting gameplay. A total of **164 source files** comprising approximately **32,100 lines of code** deliver five distinct game experiences spanning tower defense, turn-based RPG, virtual science lab, and collaborative puzzle survival genres.

This report presents each game with live screenshots, asset samples, and a detailed breakdown of the mechanics that make them both genuinely fun and pedagogically valuable.

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## Platform Overview

| Metric | Value |
|---|---|
| Total Games | 5 |
| Total Source Files | 164 |
| Total Lines of Code | ~32,100 |
| Total Question Bank | 410 questions across 10 JSON files |
| Analytics Dimensions Covered | All 6 (D1-D6) |

| Metric | Value |
|---|---|
| Engine | KAPLAY.js (browser-based, no install required) |
| Architecture | pnpm + Turborepo monorepo |

### Analytics Dimensions

| ID | Dimension | Primary Game |
|---|---|---|
| D1 | Cognitive Knowledge | All games (question-gated mechanics) |
| D2 | Behavioral Engagement | All games (progression systems) |
| D3 | Strategic Behavior & Agency | Concept Cascade, Lab Explorer |
| D4 | Social & Collaborative | Pulse Realms, Survival Equation |
| D5 | Affective & SEL | Knowledge Quest |
| D6 | Temporal & Longitudinal | All games (session-over-session tracking) |

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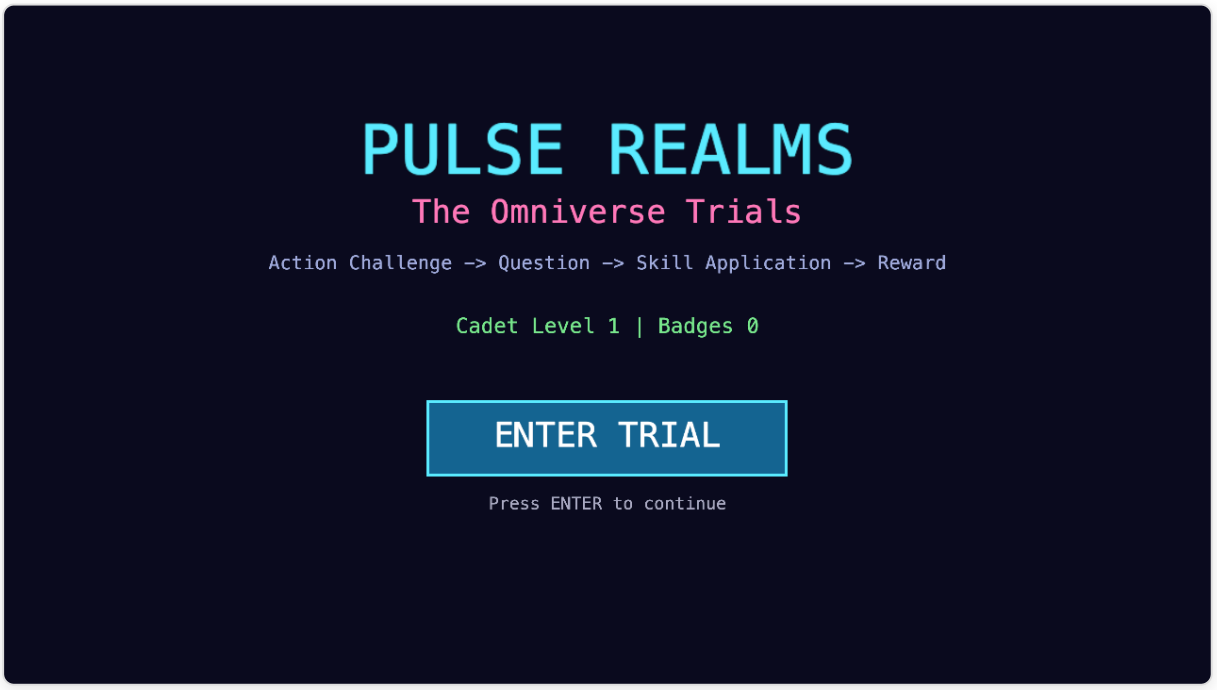
## Game 1: Pulse Realms - Team Arena

**Genre:** 3v3 Team Arena | **Subject:** Math & Science | **Primary Dimension:** D4 Social
**Status:** Complete | **Files:** 25 | **Duration:** 5 minutes per match



*Pulse Realms main menu showing the Omniverse Trials theme with progression tracking.*

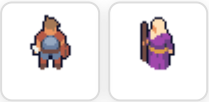
### Gameplay Mechanics

Players choose one of three combat roles (Attacker, Healer, Builder) and join a 3v3 team battle against AI opponents. Every ability -- attacks, heals, and shields -- is gated behind a multiple-choice question. Answering correctly activates the ability; answering quickly amplifies its power through a speed multiplier system.

**Key Features:** - **Role-based gameplay** with three distinct playstyles (Attacker, Healer, Builder) - **Speed multiplier system:** fast correct answers deal up to 2x damage - **Adaptive difficulty:** question difficulty adjusts based on player performance via Bayesian-inspired skill tracking - **Real-time combat** with AI teammates and opponents using state-machine behavior - **Objective control:** teams compete to hold a central capture point for score

### Assessment Integration

Every combat action generates telemetry: question accuracy, response time, role choice, team support actions (heals/shields on allies), and persistence after taking damage. The speed-accuracy profile classifies students as fluent, deliberate, guessing, or struggling.



*Sample character assets: Guardian warrior (left) and Mage spellcaster (right) representing the combat roles.*

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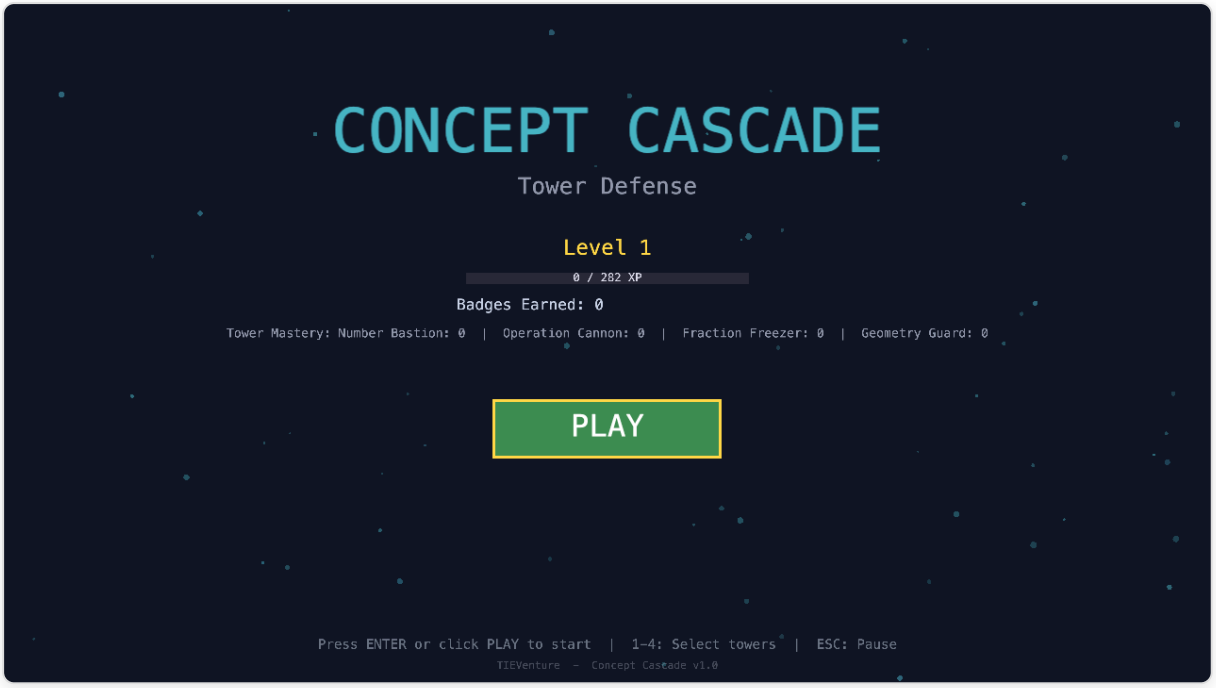
## Game 2: Concept Cascade - Tower Defense

**Genre:** Tower Defense | **Subject:** Mathematics | **Primary Dimension:** D3 Strategic
**Status:** Complete | **Files:** 32 | **Lines:** ~5,600 | **Duration:** 10-15 minutes

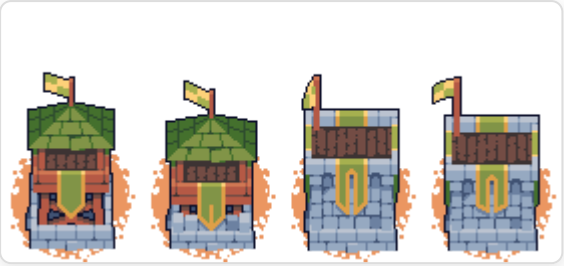



*Concept Cascade menu showing tower mastery tracking for all four tower types and the XP progression system.*

## Gameplay Mechanics

Players defend a Knowledge Core against waves of math-themed enemies by building and upgrading towers. Building a tower requires answering a math question correctly -- wrong answers refund half the cost rather than punishing harshly. The game is designed around the principle that **failure is a learning signal, not a punishment**.

**Key Features:** - **4 tower types** mapped to math knowledge components: Number Bastion (number sense), Operation Cannon (arithmetic operations), Fraction Freezer (fractions), Geometry Guard (geometry) - **Tower synergy discovery system** (inspired by Bloons TD 6): placing certain towers near each other triggers combo effects like "Chain Calculation" (2x damage) or "Shatter Shot" (3x damage on frozen enemies). Synergies are NOT documented -- players discover them through experimentation - **Risk/reward mechanics:** "Early Call" sends the next wave early for +30% bonus gold; "Risky Upgrade" attempts a harder question for a free upgrade but risks losing a tower level on failure - **8 waves + boss wave** with progressive difficulty - **5% interest** on unspent gold between waves (rewards strategic saving)



*Tower upgrade progression showing four levels of the Archer Tower, from basic wooden structure to fortified stone tower with flags -- illustrating the visual progression system in Concept Cascade.*

## Enemy Types and Assessment

Each enemy type maps to a specific mathematical knowledge component. When enemies break through defenses, the system identifies which concept the student struggles with:

| Enemy | Knowledge Component | Behavior |
|---|---|---|
| Number Sprite | Number Sense | Fast swarms that scatter when one dies |
| Operation Ogre | Operations | March in formation, pause to "flex" |
| Fraction Phantom | Fractions | Flicker in/out of visibility (hard to hit) |
| Geometry Golem | Geometry | Crack at 50% HP, split into fragments |
| Concept Dragon | Mixed (Boss) | 3 phases, spawns minions as it weakens |

*Medieval knight enemy asset representing the armored Operation Ogre -- tough, proud, and marching in formation.*

## Assessment Metrics (D3 Strategic Behavior)

- **Tower diversity:** Shannon entropy of tower type distribution (0 = one type only, 1 = perfectly diverse)
- **Strategy shifts:** detected when tower composition changes significantly between waves
- **Resource efficiency:** gold spent on towers vs. total gold earned

- **Synergy discovery count:** systems thinking indicator
- **Early call usage:** risk-taking propensity

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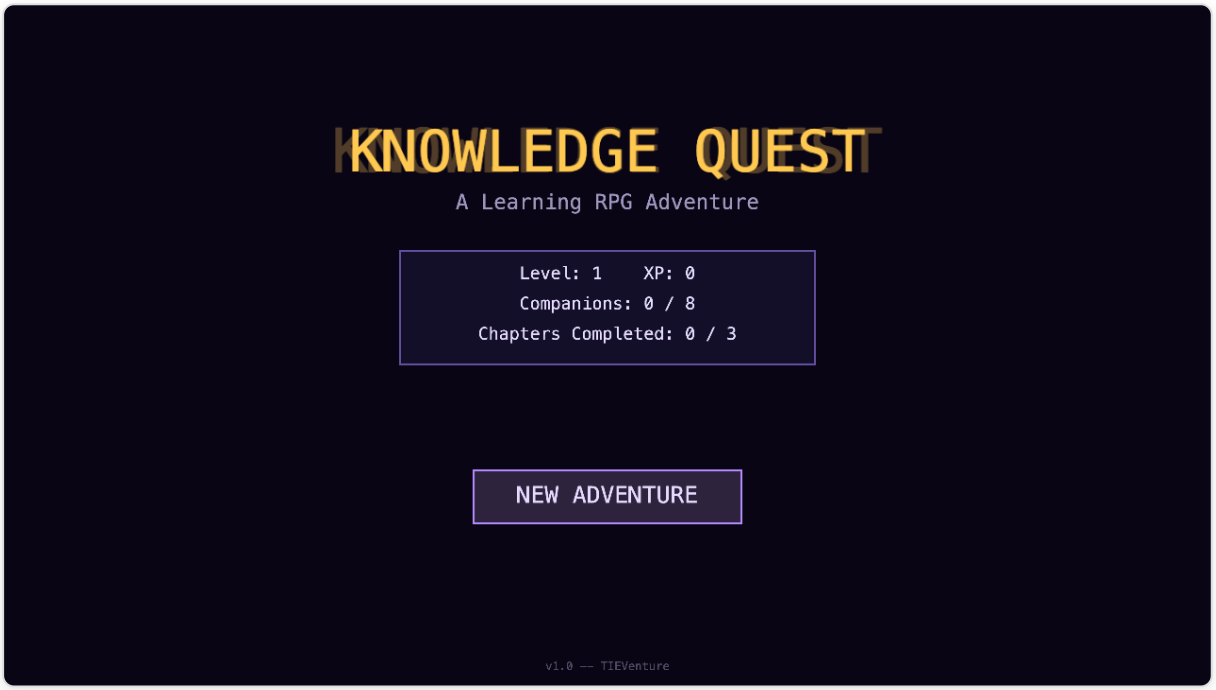
## Game 3: Knowledge Quest - Turn-Based RPG

**Genre:** Turn-Based RPG | **Subject:** Math & Science | **Primary Dimension:** D5 Affective/SEL
**Status:** Complete | **Files:** 38 | **Lines:** ~10,200 | **Duration:** 15-25 min per chapter




*Knowledge Quest menu with the golden RPG title, companion collection progress (0/8), and chapter tracking (0/3).*

### Gameplay Mechanics

A story-driven RPG where students traverse branching chapter maps (inspired by Slay the Spire), engage in turn-based combat with question-gated spells, collect Knowledge Companions (inspired by Pokemon), and make meaningful dialogue choices that affect the game world (inspired by Undertale).

**Key Features:** - **Paper Mario-style timed casts:** after answering a question correctly, a timing minigame determines spell power (PERFECT = 2x, GOOD = 1.5x, OK = 1x, MISS = 0.7x). Six timing patterns keep combat physically engaging - **8 collectible Knowledge Companions** (Pythos the Triangle, Reactia the Molecule, Algebrix the Variable, etc.) that provide passive buffs and evolve as students answer questions in their domain - **Branching**

**chapter maps** with combat, dialogue, shop, mystery, rest, and boss nodes -- visible from the start for strategic route planning - **5 enemy types with personality:** Ignorance Imps argue with each other; Confusion Crawlers shuffle your spell menu; Doubt Shades whisper discouragement; Apathy Giants fall asleep (non-violent defeat option); Boss Riddlers ask YOU riddles - **Professor Sage mentor:** a witty owl character who provides conceptual hints (not answers) with humorous commentary

**6 Social Dilemmas** across 3 chapters present genuine moral choices: - Help a lost merchant (costs time, future reward) vs. ignore vs. trade - Teach a struggling student (hard MCQ for YOU) vs. ignore vs. trade answer - Negotiate with a dragon (hardest MCQ but keep everything) vs. sacrifice vs. refuse

*Golem creature asset representing the Apathy Giant -- a slow, powerful enemy that falls asleep if you don't attack, offering a non-violent path.*

### Assessment Metrics (D5 Affective & SEL)

- **Empathy score:** ratio of prosocial to self-interest dialogue choices
- **Help-seeking pattern:** when hints are used (proactive vs. after failure vs. under time pressure)
- **Growth mindset:** chose harder paths + retried after failure + sought help then tried alone
- **Emotional regulation:** accuracy when HP is low vs. when healthy
- **Persistence:** continued engagement after combat losses

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## Game 4: Lab Explorer - Virtual Science Lab

**Genre:** Science Simulation | **Subject:** Chemistry & Physics | **Primary Dimension:** D3 Strategic
**Status:** Complete | **Files:** 31 | **Lines:** ~5,700 | **Duration:** 15-20 minutes

*Lab Explorer menu with animated bubbling beakers, experiment progress tracking, and the Discovery Journal button.*

## Gameplay Mechanics

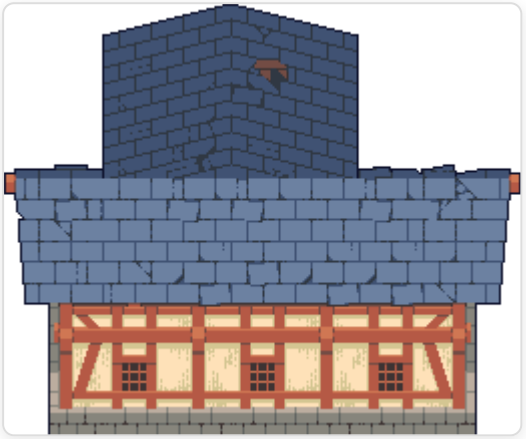
Students conduct five real science experiments through a six-phase loop: Hypothesis, Equipment Selection, Variable Manipulation, Run Experiment, Observe Results, Draw Conclusions. The game emphasizes that **failure is the best teacher** -- wrong experiments produce spectacular, entertaining animations rather than boring error messages.

### 5 Experiments:

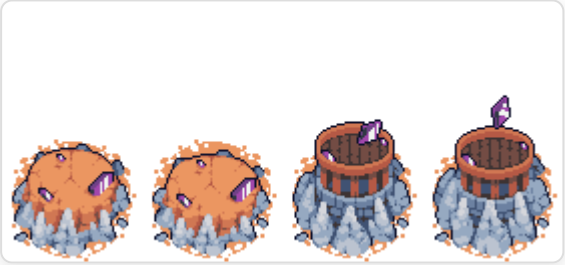
| Experiment | Subject | Key Concept |
|---|---|---|
| Acid-Base Balancing | Chemistry | pH neutralization |
| Density Detective | Physics | Mass/volume relationships |
| Simple Circuits | Physics | Ohm's law (V = IR) |
| Pendulum Period | Physics | Only length affects period (not mass!) |
| Heat Transfer | Physics/Chemistry | Insulation and thermal conductivity |

**Key Features:** - **Spectacular failure states** (inspired by Kerbal Space Program): foam eruptions when mixing too much acid, sparks flying from short circuits, pendulum strings breaking, beakers dissolving -- each failure is unique, animated, and teaches something - **Disaster Gallery:** collectible failure achievements (encourages experimentation!) - **Real-**

**time visual feedback:** solution colors change smoothly as variables adjust, pendulums swing with accurate physics, circuit diagrams light up as current flows - **Discovery Journal:** 15 hidden findings across all experiments (e.g., "Does mass affect pendulum period?" -- discovering it doesn't earns "Galileo's Insight") - **Professor Challenge mode:** harder versions of completed experiments for replay value - **Multiple valid solutions:** density can be measured via water displacement OR comparison to known materials



*Workshop building asset representing the science laboratory environment -- a detailed medieval workshop with chimney and timber framing.*



*Crystal structure progression showing four stages of magical experimentation -- representing the visual evolution of experiments from simple to complex.*

## Assessment Metrics (D3 Strategic Behavior)

- **Systematic experimentation:** did the student change only one variable at a time? (process mining from full action log)
- **Equipment selection quality:** chose correct tools from a shelf of options
- **Exploration breadth:** variables tried beyond the minimum required
- **Self-correction rate:** adjusted approach after unexpected results
- **Measurement accuracy:** how close results are to correct values

## Game 5: Survival Equation - Collaborative Puzzle Survival

**Genre:** Cooperative Puzzle | **Subject:** Applied Math & Science | **Primary Dimension:** D4 Social
**Status:** Complete | **Files:** 38 | **Lines:** ~6,700 | **Duration:** 15-20 min per scenario

*Survival Equation menu with the "Collaborate. Calculate. Survive." tagline and the core gameplay loop description.*

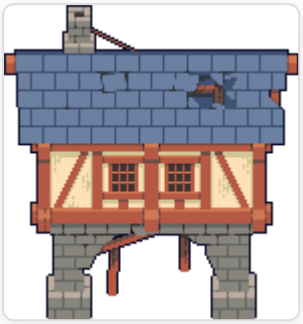
### Gameplay Mechanics

A team of four specialists -- Engineer, Scientist, Medic, Navigator -- are stranded in a hostile environment. Each has exclusive information that others cannot see. Puzzles are literally unsolvable alone; students must communicate with AI teammates to share data and solve survival challenges. Inspired by Keep Talking and Nobody Explodes (information asymmetry) and Overcooked (escalating chaos).

**3 Scenarios:** - **Desert Island:** water purification, shelter construction, rescue signal - **Space Station:** oxygen recycling, hull repair, escape pod launch - **Underwater Base:** pressure seals, desalination, emergency buoy

**Key Features:** - **Information asymmetry IS the game:** the Engineer sees material specs, the Scientist sees formulas, the Medic sees health requirements, the Navigator sees terrain maps -- no single role has enough information to solve any puzzle - **AI partners with personality:** Raza (Engineer, confident/overestimates), Juno (Scientist, precise/wordy), Kit (Medic, nervous/caring), Navi (Navigator, adventurous/risk-taking) - **Varied mini-puzzles:**

drag-and-drop filter layers, beam placement physics, circuit wiring, math allocation sliders, map navigation - **Escalating daily events:** storms, teammate sickness, rival camps, final rescue countdown - **Resource management:** shared food/water/materials with daily consumption -- allocation decisions are tracked for fairness - **Day countdown timer:** sky darkens as time runs low, creating natural dramatic tension



*Elevated survival shelter with reinforced stone pillars and shingled roof -- representing the shelter construction puzzle in the Desert Island scenario.*

*Terrain platform asset showing the rocky survival environment where teams must build and survive.*

### Assessment Metrics (D4 Social & Collaborative)

- **Communication quality:** on-task message ratio, information sharing frequency
- **Team contribution equity:** Gini coefficient measuring how evenly work is distributed
- **Leadership patterns:** proposals made, questions asked, directions given
- **Information sharing:** proactive vs. reactive information exchange
- **Resource fairness:** equal distribution vs. hoarding behavior
- **Role adoption:** leader, supporter, executor, or observer classification

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## Technical Architecture

### Shared Infrastructure

All five games share three core systems copied from Pulse Realms:

1. **Telemetry System** ( `telemetry.js` ): Event capture with localStorage backup and REST API batch flush every 10 seconds. Supports offline play with automatic sync when connectivity returns.

2. **Question Engine** ( `questionEngine.js` ): Adaptive difficulty using skill rating (1-5), streak tracking (3 correct = difficulty up, 2 wrong = difficulty down), and subject-specific question banks loaded from JSON files.

3. **Progression System** ( `progression.js` ): XP, levels, and badges with per-game customization. XP formula: `baseXP * difficultyMultiplier * speedBonus * correctnessFactor` .

### Monorepo Structure

```
apps/games/
  pulse-realms/      25 files   (3v3 Team Arena)
  concept-cascade/   32 files   (Tower Defense)
  knowledge-quest/   38 files   (Turn-Based RPG)
  lab-explorer/      31 files   (Virtual Science Lab)
  survival-equation/ 38 files   (Collaborative Puzzle)
```

Each game follows an identical internal architecture:

```
src/
  config/       Game constants, entity definitions
  data/         Questions (JSON), story scripts, maps
  systems/      Game logic (state, combat, assessment)
  components/   KAPLAY.js UI components and entities
  scenes/       Scene registration and game flow
```

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## Question Bank Summary

| Game | Files | Total Questions | Subjects |
|---|---|---|---|
| Concept Cascade | 4 | 60 | Number sense, Operations, Fractions, Geometry |
| Knowledge Quest | 2 | 150 | Math (75), Science (75) |

| Game | Files | Total Questions | Subjects |
|---|---|---|---|
| Lab Explorer | 2 | 100 | Chemistry (50), Physics (50) |
| Survival Equation | 2 | 100 | Applied Math (50), Applied Science (50) |
| **Total** | **10** | **410** | |

All questions are themed to their game context -- tower defense math problems, RPG spell-casting challenges, lab experiment questions, and survival-scenario word problems. Each file contains questions across difficulty levels 1-5 (easiest to hardest).

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## Evidence-Centered Design Coverage

Every game has a complete ECD mapping document specifying: - **Competency Model:** what knowledge, skills, and attributes are measured - **Evidence Model:** telemetry events and evidence rules connecting behavior to competencies - **Task Model:** game mechanics calibrated to specific difficulty and skill targets - **Assembly Model:** mastery thresholds and adaptive difficulty parameters

### Coverage Matrix

| | D1 Cognitive | D2 Engagement | D3 Strategic | D4 Social | D5 Affective | D6 Temporal |
|---|---|---|---|---|---|---|
| Pulse Realms | Strong | Strong | Medium | **Primary** | Medium | Medium |
| Concept Cascade | Strong | Strong | **Primary** | Weak | Medium | Strong |
| Knowledge Quest | Strong | Strong | Strong | Medium | **Primary** | Strong |
| Lab Explorer | Strong | Medium | **Primary** | Medium | Medium | Strong |
| Survival Equation | Strong | Strong | Strong | **Primary** | Strong | Strong |

Every analytics dimension has at least one game as its primary source, ensuring comprehensive coverage of student competencies.

## Next Steps

1. **Browser Testing & Debugging:** Systematic playtesting of all four new games
2. **Pilot Study Design:** Prepare for classroom testing in Saudi Arabian K-12 schools
3. **Teacher Dashboard Integration:** Connect game telemetry to real-time analytics dashboards
4. **Tablet Adaptation:** Test Pulse Realms on iPad/Android for cross-platform assessment equivalence
5. **AI Question Generation:** Implement the teacher-upload-to-MCQ pipeline
6. **SpacetimeDB Integration:** Add real-time multiplayer for Pulse Realms and Survival Equation